Factors and Multiples

Factors make the number. Multiples come from counting by it.

Every rectangle for 12



1 × 12 2 × 6 3 × 4

Factor pairs of 12: 1 and 12, 2 and 6, 3 and 4.

Factors of 12: 1, 2, 3, 4, 6, 12

Multiples: count by the number

Multiples of 4: 4, 8, 12, 16, 20, …

12 is a multiple of 4, and 4 is a factor of 12.

Prime or composite?

7 has only one rectangle: 1 × 7. It is prime.



8 has more: 2 × 4 and 1 × 8. It is composite.

Your turn!

$$15 = 1 \times \boxed{15} \text{ and } 3 \times \boxed{5}$$

For teachers

Goal: factor pairs, multiples, prime and composite for numbers to 100 (4.OA.4). Say “Can you make a rectangle with this many dots? How many rows?” Start at 1 and stop when the pairs turn around.

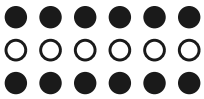
Factors and Multiples: Practice

A factor divides the number with nothing left over.

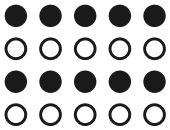
Part 1 Rectangles How many in each row?

1) 

$2 \times \boxed{9} = 18$

2) 

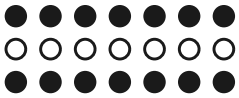
$3 \times \boxed{6} = 18$

3) 

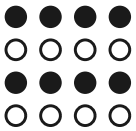
$4 \times \boxed{5} = 20$

4) 

$2 \times \boxed{8} = 16$

5) 

$3 \times \boxed{7} = 21$

6) 

$4 \times \boxed{4} = 16$

Part 2 Factor pairs Start with 1. Stop when the pairs turn around.

7) $18 = 1 \times \boxed{18}$ and $2 \times \boxed{9}$ and $3 \times \boxed{6}$

Factors and Multiples: Practice (continued)

Part 2 Factor pairs (continued)

8) $20 = 1 \times \boxed{20}$ and $2 \times \boxed{10}$ and $4 \times \boxed{5}$

9) $28 = 1 \times \boxed{28}$ and $2 \times \boxed{14}$ and $4 \times \boxed{7}$

10) $16 = 1 \times \boxed{16}$ and $2 \times \boxed{8}$ and $4 \times \boxed{4}$

11) $30 = 1 \times \boxed{30}$ and $2 \times \boxed{15}$ and $3 \times \boxed{10}$ and $5 \times \boxed{6}$

12) $21 = 1 \times \boxed{21}$ and $3 \times \boxed{7}$

13) $36 = 1 \times \boxed{36}$ and $2 \times \boxed{18}$ and $3 \times \boxed{12}$ and $4 \times \boxed{9}$ and $6 \times \boxed{6}$

14) $27 = 1 \times \boxed{27}$ and $3 \times \boxed{9}$

Part 3 Is it a factor? Circle yes or no.

15) Is 6 a factor of 42? (yes) no

16) Is 4 a factor of 30? yes (no)

17) Is 9 a factor of 48? yes (no)

18) Is 7 a factor of 56? (yes) no

Factors and Multiples: Practice (continued)

Part 3 Is it a factor? (continued)

19) **Is 8 a factor of 72?** (yes) no

20) **Is 3 a factor of 51?** (yes) no

21) **Is 5 a factor of 32?** yes (no)

22) **Is 6 a factor of 50?** yes (no)

23) **Is 4 a factor of 26?** yes (no)

24) **Is 7 a factor of 84?** (yes) no

Part 4 All the factors No help this time. List them from smallest to biggest.

25) **Factors of 24:** 1 2 3 4 6 8 12 24

26) **Factors of 40:** 1 2 4 5 8 10 20 40

27) **Factors of 32:** 1 2 4 8 16 32

28) **Factors of 45:** 1 3 5 9 15 45

29) **Factors of 50:** 1 2 5 10 25 50

30) **Factors of 60:** 1 2 3 4 5 6 10 12 15 20 30 60

Factors and Multiples: Practice (continued)

Part 5 Multiples No help this time. Count by the number.

31) **Multiples of 6: 6 12** 18 24 30

32) **Multiples of 9: 9 18** 27 36 45

33) **Multiples of 7: 7 14** 21 28 35

34) **Multiples of 12: 12 24** 36 48 60

35) **Multiples of 8: 8 16** 24 32 40

36) **Multiples of 11: 11 22** 33 44 55

37) **Multiples of 15: 15 30** 45 60 75

38) **Multiples of 25: 25 50** 75 100 125

Part 6 Is it a multiple? Circle yes or no.

39) **Is 63 a multiple of 9?** (yes) no

40) **Is 44 a multiple of 6?** yes (no)

41) **Is 35 a multiple of 5?** (yes) no

42) **Is 81 a multiple of 3?** (yes) no

Factors and Multiples: Practice (continued)

Part 6 Is it a multiple? (continued)

43) **Is 52 a multiple of 8?** yes (no)

44) **Is 70 a multiple of 4?** yes (no)

45) **Is 96 a multiple of 8?** (yes) no

46) **Is 58 a multiple of 7?** yes (no)

47) **Is 74 a multiple of 9?** yes (no)

48) **Is 90 a multiple of 6?** (yes) no

Part 7 Prime or composite? Circle one.

49) **17** (prime) composite

50) **21** prime (composite)

51) **9** prime (composite)

52) **13** (prime) composite

53) **2** (prime) composite

54) **27** prime (composite)

Factors and Multiples: Practice (continued)

Part 7 Prime or composite? (continued)

55) **39** prime (composite)

56) **11** (prime) composite

57) **31** (prime) composite

58) **29** (prime) composite

59) **51** prime (composite)

60) **49** prime (composite)

Part 8 Is it a factor? Circle yes or no.

61) **Is 2 a factor of 41?** yes (no)

62) **Is 9 a factor of 37?** yes (no)

63) **Is 9 a factor of 90?** (yes) no

64) **Is 6 a factor of 66?** (yes) no

65) **Is 8 a factor of 21?** yes (no)

66) **Is 8 a factor of 16?** (yes) no

Factors and Multiples: Practice (continued)

Part 8 Is it a factor? (continued)

67) **Is 6 a factor of 24?** (yes) no

68) **Is 4 a factor of 69?** yes (no)

69) **Is 6 a factor of 44?** yes (no)

70) **Is 8 a factor of 88?** (yes) no

71) **Is 2 a factor of 86?** (yes) no

72) **Is 3 a factor of 18?** (yes) no

73) **Is 9 a factor of 44?** yes (no)

74) **Is 6 a factor of 18?** (yes) no

75) **Is 2 a factor of 70?** (yes) no

76) **Is 7 a factor of 66?** yes (no)

Factors and Multiples: Practice (continued)

Part 9 Prime or composite?

77) **79** (prime) composite

78) **42** prime (composite)

79) **85** prime (composite)

80) **55** prime (composite)

81) **26** prime (composite)

82) **93** prime (composite)

83) **48** prime (composite)

84) **74** prime (composite)

85) **23** (prime) composite

86) **52** prime (composite)

87) **83** (prime) composite

88) **32** prime (composite)

Factors and Multiples: Practice (continued)

Part 9 Prime or composite? (continued)

89) **24** prime (composite)

90) **88** prime (composite)

Part 10 Remember? Problems from earlier packets.

91) **846 ÷ 3 =** 282

92) **31 × 22 =** 682

93) **4,568 to the nearest ten =** 4,570

94) $\frac{2}{3}$ **of 15 =** 10

95) **2,705 + 4,198 =** 6,903

96) $\frac{5}{6}$ **of 30 =** 25

Part 11 Word problems Draw the rectangles.

97) A teacher wants 24 desks in equal rows. Will rows of 5 work, or rows of 6?

Number sentence 24 = 4 × 6, 24 ÷ 5 = 4 R 4

Answer rows of 6

Draw or show your work

98) Ana packs 36 cookies into bags with the same number in each. Should she put 7 or 9 in a bag?

Number sentence 36 = 4 × 9

Answer 9 in a bag

Draw or show your work

99) Buses come every 6 minutes. Does a bus come at 42 minutes after the hour?

Number sentence 42 = 7 × 6, a multiple of 6

Answer yes

Draw or show your work

Factors and Multiples: Practice (continued)

Part 11 Word problems (continued)

100) Leo has 29 tiles. Can he make a rectangle with more than one row?

Number sentence **29 is prime: only 1 × 29**

Answer **no**

Draw or show your work

101) Challenge: What is the smallest number that is a multiple of both 4 and 6?

Number sentence **4, 8, 12 and 6, 12: both have 12**

Answer **12**

Draw or show your work

Check-In: Factors and Multiples

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Factor pairs

1) $48 = 1 \times \boxed{48}$ and $2 \times \boxed{24}$ and $3 \times \boxed{16}$ and $4 \times \boxed{12}$ and $6 \times \boxed{8}$

2) $40 = 1 \times \boxed{40}$ and $2 \times \boxed{20}$ and $4 \times \boxed{10}$ and $5 \times \boxed{8}$

Part 2 Is it a factor? Circle yes or no.

3) Is 3 a factor of 60? (yes) no

4) Is 4 a factor of 90? yes (no)

Part 3 Is it a multiple? Circle yes or no.

5) Is 48 a multiple of 6? (yes) no

6) Is 62 a multiple of 6? yes (no)

Part 4 Prime or composite? Circle one.

7) 53 (prime) composite

8) 46 prime (composite)

9) 58 prime (composite)